

App Dev Project Report

1. Student Details

Name: Nicky Yadav

Roll Number: 24f2008109

Email: 24f2008109@ds.study.iitm.ac.in

Program: IIT Madras BS in Data Science and Applications

2. Project Details

Project Title

PlaceHub - Placement Portal Application V2

Problem Statement

Institutes often manage campus placement activities using spreadsheets, emails, and manual coordination. This makes it difficult to efficiently manage company registrations, approvals, placement drives, student applications, and recruitment status.

PlaceHub provides a centralized web application where the Institute Placement Cell, Companies, and Students can interact through role-based workflows.

Approach

The application uses **Flask** to provide REST APIs and **Vue.js** for the user interface.

SQLite is used for persistent data storage, while **JWT** is used for authentication and role-based decorators are used to protect Admin, Company, and Student resources.

Redis and Celery are used for asynchronous background jobs such as student application CSV export and monthly placement activity report generation.

Bootstrap and **Bootstrap Icons** are used to provide a responsive and consistent user interface.

3. AI/LLM Declaration

AI/LLM tools were used as development assistance for understanding concepts, debugging errors, improving code structure, and preparing documentation.

The project implementation, integration, testing, and final decisions were carried out and verified manually by the student.

4. Technologies and Frameworks Used

Technology / Library	Purpose
Flask	Backend REST API framework
Vue.js	Frontend user interface
Bootstrap	Responsive UI and styling
Bootstrap Icons	Icons used throughout the frontend
SQLite	Application database
SQLAlchemy	ORM and database operations
Flask-JWT-Extended	JWT-based authentication
Flask-CORS	Cross-origin API communication
Axios	Frontend-to-backend API communication
Vue Router	Client-side routing
Pinia	Frontend authentication and state management
Redis	Celery broker and result backend

Technology / Library	Purpose
Celery	Asynchronous background job processing

5. Database Schema / ER Diagram

The application contains the following main database models:

User

Stores common user account information.

Fields include:

- ID
- Name
- Email
- Password Hash
- Role
- Blocked Status
- Created At
- Updated At

The supported roles are: Admin, Company, Student.

CompanyProfile

Stores information related to registered companies.

Fields include:

- Company ID
- User ID
- Company Name
- Website
- Location
- Description
- Approval Status
- Created At
- Updated At

StudentProfile

Stores information related to registered students.

Fields include:

- Student ID
- User ID
- College

- CGPA
- Skills
- Resume
- Created At
- Updated At

PlacementDrive

Stores placement drive information created by companies.

Fields include:

- Drive ID
- Company ID
- Job Title
- Job Description
- Package
- Location
- Application Deadline
- Eligibility Criteria
- Created At
- Updated At

Application

Stores information about students applying to placement drives.

Fields include:

- Application ID
- Student ID
- Drive ID
- Application Status
- Applied At
- Created At
- Updated At

Application statuses include: Applied, In Touch, Shortlisted, Rejected.

Database Relationships

User

```
|
|---- One-to-One ---- CompanyProfile
|
|---- One-to-One ---- StudentProfile
```

CompanyProfile

```
|
|---- One-to-Many ---- PlacementDrive
```

StudentProfile
|
|---- One-to-Many ---- Application

PlacementDrive
|
|---- One-to-Many ---- Application

A User can have either a CompanyProfile or StudentProfile depending on the user's role.
A Company can create multiple Placement Drives.
A Student can submit multiple Applications for different Placement Drives.
A Placement Drive can receive Applications from multiple Students.

6. API Resource Endpoints

The application follows a REST API architecture.

Authentication APIs

Endpoint	Method	Description
/api/auth/*	POST / GET	Registration, login, and authenticated user details

Admin APIs

Endpoint	Method	Description
/api/admin/dashboard	GET	Fetch Admin dashboard statistics
/api/admin/users	GET	View registered users
/api/admin/companies	GET	View registered companies

Endpoint	Method	Description
/api/admin/companies/<id>/approval	PATCH	Approve or reject a company
/api/admin/students	GET	View registered students
/api/admin/drives	GET	View placement drives
/api/admin/applications	GET	View all student applications
/api/admin/generate-monthly-report	POST	Start asynchronous monthly report generation

Student APIs

Endpoint	Method	Description
/api/student/dashboard	GET	Fetch Student dashboard
/api/student/profile	GET	Fetch Student profile
/api/student/profile	PUT	Update Student profile
/api/student/drives	GET	View available placement drives
/api/student/apply/<drive_id>	POST	Apply to a placement drive

Endpoint	Method	Description
/api/student/applications	GET	View Student applications
/api/student/export-applications	POST	Start asynchronous CSV export
/api/student/export-status/<task_id>	GET	Check CSV export task status
/api/student/download-export/<filename>	GET	Download generated CSV file

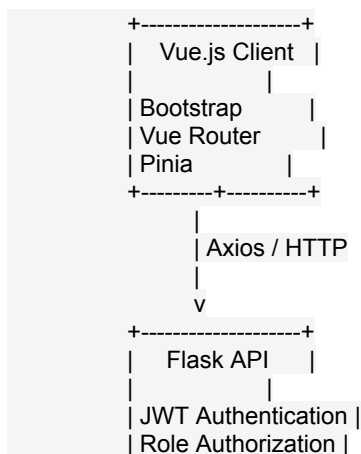
Company APIs

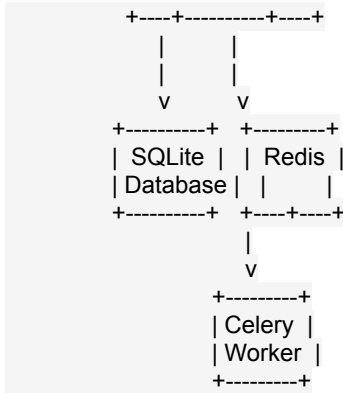
Company APIs provide functionality to:

- View Company dashboard
- View and update Company profile
- Create Placement Drives
- View created Placement Drives
- View Student Applications
- Update Application Status

7. Architecture and Features

Architecture Overview





The Vue.js frontend communicates with Flask REST APIs using Axios.

JWT tokens are used to authenticate protected API requests.

SQLAlchemy manages database operations with SQLite.

Redis is used as infrastructure for Celery background jobs.

Celery workers execute long-running jobs separately from normal HTTP requests.

Implemented Features

Admin Features

- Pre-existing Admin account
- Secure Admin login
- Admin dashboard
- View system statistics
- View registered users
- View registered students
- View registered companies
- Approve company registrations
- Reject company registrations
- Block users
- Unblock users
- Delete users
- View placement drives
- Delete placement drives
- View student applications
- Generate Monthly Placement Activity Report
- Download generated HTML reports

Company Features

- Company registration
- Company login
- Company profile management
- Admin approval workflow
- Company dashboard

- Create placement drives
- View created placement drives
- View student applications
- View Student College and CGPA
- Update application status (Applied, In Touch, Shortlisted, Rejected)

Student Features

- Student registration
- Student login
- Student dashboard
- View Student profile
- Update Student profile
- View available placement drives
- Apply for placement drives
- Prevent duplicate applications
- View submitted applications
- Track application status
- View application history
- Export application history as CSV
- Download generated CSV file

Background Jobs

PlaceHub uses **Redis and Celery** for asynchronous background processing.

Student Application CSV Export

Student Clicks Export -> Flask API Creates Celery Task -> Task Added to Redis Queue
 -> Celery Worker Processes Task -> CSV File Generated -> Frontend Checks Task Status
 -> CSV File Downloaded

Admin Monthly Activity Report

Admin Clicks Generate Report -> Flask API Creates Celery Task -> Task Added to Redis Queue
 -> Celery Worker Generates Report -> HTML Report Saved -> Frontend Checks Task Status
 -> Admin Downloads Report

Security Features

- JWT-based authentication
- Role-based authorization
- Password hashing
- Protected Admin APIs

- Protected Company APIs
- Protected Student APIs
- Custom role-based decorators
- User blocking functionality
- Company approval system

8. Video Presentation

Drive Link: [📺 ProPlaced](#)